

TPC-405: An Inverse-Height Bound in the C1 Local-Normalization Proxy

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Abstract

TPC-404 identified the exact local geometry of a CRT-generated signed deletion profile but tested only four finite configurations. We prove the next scoped statement: under explicit integer window and shell hypotheses, the locally normalized adjacent entry of the selected-prime CRT proxy is at most $t_1/(a_{\min}\sqrt{S_0S_1}) \leq 4/(a_{\min}H) \leq 4/H$. The proof is a Cauchy–Schwarz estimate combined with the two local energy identities. An exact rational certificate independently audits five heights and four multiplicities, while its decimal scale ladder is recorded only as a finite numerical observation. This is a uniform bound for one synthetic proxy entry, not a bound on the complete operator and not an arithmetic or twin-prime theorem.

1 Scope and model

Let H, N, Q be integers with $H \geq 1$, $N \geq H + 2$, and $Q > N$. Let $p_0 < \dots < p_{2m-1}$ be distinct primes in $(Q, 2Q]$, with $m \geq 1$. Choose a CRT solution o satisfying

$$o \equiv 0 \pmod{p_i} \quad (i \text{ even}), \quad o \equiv -N \pmod{p_i} \quad (i \text{ odd}).$$

The origin may be chosen above any prescribed lower bound. The half-open window is $I_o = \{o, \dots, o + N - 1\}$.

Set

$$t_d = \frac{H^2}{H^2 + d^2}, \quad S_0 = \sum_{d=1}^{N-1} t_d^2, \quad S_1 = \sum_{d=1}^{N-2} t_d^2 + t_1^2,$$

and $a_i = p_i^3/[Q^2(p_i - 1)]$. Write

$$P_- = \sum_{i \text{ odd}} a_i, \quad V_- = \sum_{i \text{ odd}} a_i^2, \quad V_+ = \sum_{i \text{ even}} a_i^2.$$

The local selected-prime proxy inherited from TPC-404 has

$$G_0 = V_- S_0, \quad G_1 = V_- S_1 + V_+(S_1 - t_1^2), \quad M = t_1 P_-.$$

Here H is the kernel height of the proxy; it is not the physical h_0 .

2 Uniform adjacent-entry theorem

Theorem. Under the hypotheses above, put $a_{\min} = \min_i a_i$ and $z = M/\sqrt{G_0 G_1}$. Then

$$0 \leq z \leq \frac{t_1}{a_{\min} \sqrt{S_0 S_1}} \leq \frac{4}{a_{\min} H} \leq \frac{4}{H}. \quad (1)$$

Proof. Every quantity defining the amplitudes and kernel is positive. Moreover

$$S_1 - t_1^2 = \sum_{d=1}^{N-2} t_d^2 \geq 0,$$

so $G_1 \geq V_- S_1$. Cauchy–Schwarz and the m odd indices give $P_-^2 \leq m V_-$ and $V_- \geq m a_{\min}^2$. Hence

$$z^2 = \frac{t_1^2 P_-^2}{G_0 G_1} \leq \frac{t_1^2 m V_-}{(V_- S_0)(V_- S_1)} = \frac{t_1^2 m}{V_- S_0 S_1} \leq \frac{t_1^2}{a_{\min}^2 S_0 S_1}. \quad (2)$$

For $1 \leq d \leq H$ we have $t_d \geq 1/2$. Since $N \geq H + 2$, these terms occur in both S_0 and S_1 , giving $S_0, S_1 \geq H/4$. Finally,

$$a_i = \left(\frac{p_i}{Q} \right)^2 \frac{p_i}{p_i - 1} > 1$$

because $p_i > Q$. Taking square roots in (2) proves (1). $\square$

3 Exact scale ladder

The certificate uses $Q = 8192$, the first eight primes in $(Q, 2Q]$, and the five heights $H = 16, 32, 66, 128, 256$ with $N = 4H$. For every height it audits $m = 1, 2, 3, 4$, giving 20 exact rational cases. The CRT origin is reconstructed above 10^6 and the independent checker recomputes the literal masked row energies, rather than importing the producer’s local formulas.

H	$m = 1$ observation	$m = 4$ observation
16	0.057270350454587	0.057018931354529
32	0.028416761665929	0.028292078588366
66	0.013715058563593	0.013654898530787
128	0.007056083219706	0.007025136252874
256	0.003523723340942	0.003508269858201

The table contains float64 square-root evaluations of exact rational normalized squares and is therefore a *NUMERICAL OBSERVATION*. It illustrates the inverse-height scale, whereas the theorem is the symbolic inequality (1).

4 Route boundary

The theorem is a real extension of TPC-404, but its object is deliberately narrow. It bounds one adjacent entry of a selected-prime synthetic CRT proxy. It does not control the complete prime shell, unselected primes, arbitrary origins, the arithmetic sign source, or the physical h_0 . Accordingly it does not pay an arithmetic L^2 estimate, fixed-power $1/400$ credit, Route A, Route B, or a twin-prime conclusion. In particular, an upper bound for this entry is not a full operator-norm estimate.

uniform proxy-entry bound	PROVED_UNIFORM
finite rational certificate	PROVED_EXACT_FINITE
scale-ladder decimals	NUMERICAL_OBSERVATION
full normalized operator theorem	OPEN
arithmetic advance / fixed-power credit	NO / 0
Route-A / Route-B / twin-prime result	OPEN / OPEN / NONE

The next scoped question is TEST_C1_LOCAL_NORMALIZATION_COMPLETE_SHELL_ENTRY_BOUNDARY.

Reproduction

The project contains README, code, experiments, results, notes, proof package, route evaluation, and `paper/paper.pdf`. The README lists the exact producer, independent literal replay, and seven-mutation stress commands.